

Graphing Quadratic Functions

A **quadratic function** is a second-degree polynomial function.

Standard Form of a Quadratic Function

The standard form of a quadratic function is $f(x) = ax^2 + bx + c$, where $a \neq 0$. The graph of a quadratic function is called a **parabola**.

One way to graph a quadratic function is to create a table of ordered pairs, plot the points, and then draw a smooth curve through those points. As you study the following examples, recall that y and $f(x)$ are used interchangeably.

Using a Table to Graph a Quadratic Function

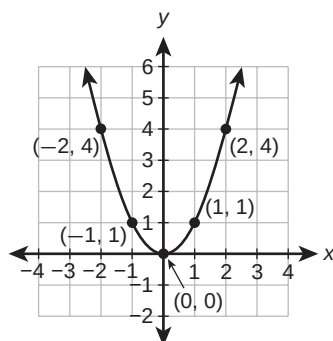
EXAMPLE 1

Graph the function.

A $y = x^2$

SOLUTION

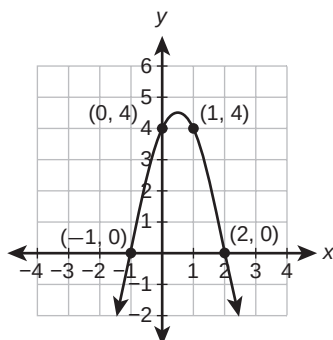
x	-2	-1	0	1	2
y	4	1	0	1	4



B $y = -2x^2 + 2x + 4$

SOLUTION

x	-2	-1	0	1	2
y	-8	0	4	4	0



The point $(-2, -8)$ is not shown.

Graphing Quadratic Functions in Standard Form

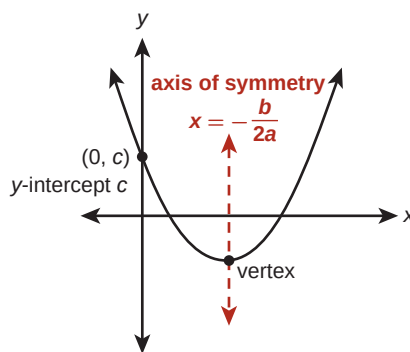
All quadratic function graphs have in common the following characteristics, which can be used to create accurate graphs.

Properties of Quadratic Function Graphs

The graph of $f(x) = ax^2 + bx + c$ is a parabola with these characteristics:

- It opens up when $a > 0$; it opens down when $a < 0$.
- It has y-intercept c , so $(0, c)$ is a point on the graph.
- It has a vertex with x-coordinate $-\frac{b}{2a}$.
- It has an axis of symmetry with equation $x = -\frac{b}{2a}$.

The axis of symmetry is the vertical line through the vertex. It separates the graph into two halves that are reflections (mirror images) of each other.



► **Think About It** The axis of symmetry is dashed because it is not part of the graph of the function.

EXAMPLE 2

Graph $y = 3x^2 - 12x + 6$.

SOLUTION

First identify the coefficients a , b , and c .

$$y = \underbrace{3x^2 - 12x + 6}_{a = 3 \quad b = -12 \quad c = 6}$$

The leading coefficient a is positive, so the parabola opens up, not down.

The x -coordinate of the vertex is $-\frac{b}{2a} = -\frac{(-12)}{2 \cdot 3} = 2$. To find the y -coordinate of the vertex, substitute 2 for x .

$$y = 3x^2 - 12x + 6 = 3 \cdot 2^2 - 12 \cdot 2 + 6 = -6$$

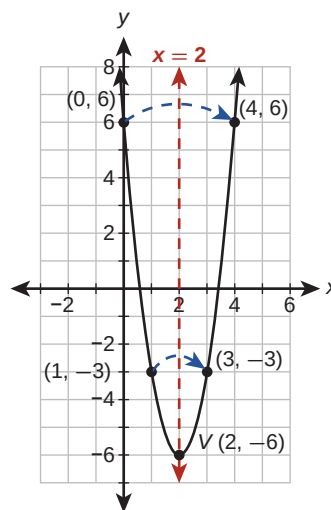
The vertex, V , is $(2, -6)$, and the equation of the axis of symmetry is $x = 2$.

The y -intercept is $c = 6$, so $(0, 6)$ is a point on the graph.

The reflection image of $(0, 6)$ over the axis of symmetry is $(4, 6)$, which is also on the graph.

Choose any value for x and substitute it to find one more point on the parabola. If $x = 1$, then $y = 3 \cdot 1^2 - 12 \cdot 1 + 6 = -3$. So $(1, -3)$ is on the graph, and its reflection image $(3, -3)$ is also on the graph.

Draw a smooth curve through all five points, including the vertex.



Determining the Number of Zeros of a Quadratic Function

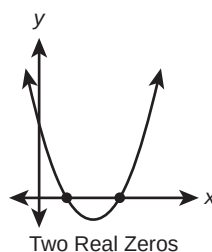
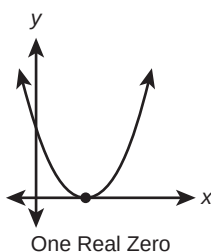
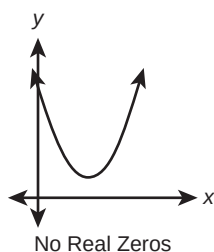
Using properties of quadratic functions can help you determine how many times the graph of a given function crosses the x -axis.

Zeros of a Polynomial Function

The **zeros of a polynomial function** $f(x)$ are the **roots** (solutions) of the equation $f(x) = 0$. The real zeros are the x -intercepts of the graph of $f(x)$.

Number of Real Zeros of a Quadratic Function

A quadratic function can have no, one, or two real zeros.



EXAMPLE 3

Determine the number of real zeros of the quadratic function.

A $y = x^2 + 6x + 9$

SOLUTION

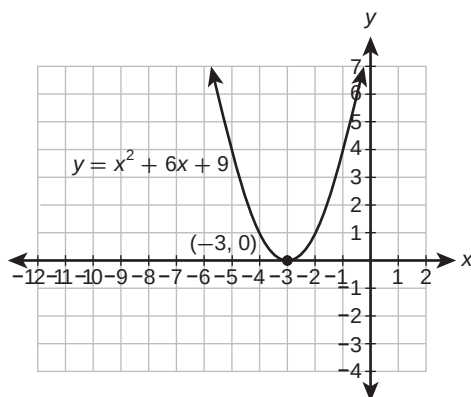
$a = 1$, $b = 6$, and $c = 9$. Since $a > 0$, the parabola opens up.

Use a and b to find the x -coordinate of the vertex, and then substitute to find the y -coordinate.

$$x = -\frac{b}{2a} = -\frac{6}{2 \cdot 1} = -3$$

$$y = x^2 + 6x + 9 = (-3)^2 + 6 \cdot (-3) + 9 = 9 - 18 + 9 = 0$$

The vertex is $(-3, 0)$. The vertex is a zero of the function and because the parabola opens up, the vertex is also the lowest point on the graph. Therefore, it is the only real zero. Sketch the graph to verify.



B $y = x^2 + 2$

SOLUTION

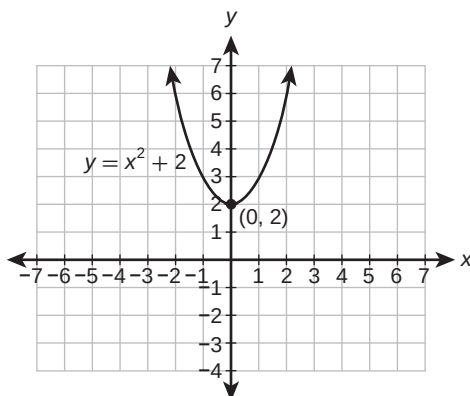
$a = 1$, $b = 0$, and $c = 2$. Since $a > 0$, the parabola opens up.

Use a and b to find the x -coordinate of the vertex, and then substitute to find the y -coordinate.

$$x = -\frac{b}{2a} = -\frac{0}{2 \cdot 1} = 0$$

$$y = x^2 + 2 = 0^2 + 2 = 2$$

The vertex is $(0, 2)$. Since the parabola opens up, the vertex is the lowest point on the graph. Therefore, the function has no real zeros. Sketch the graph to verify.



C $y = -x^2 + 4x - 1$

SOLUTION

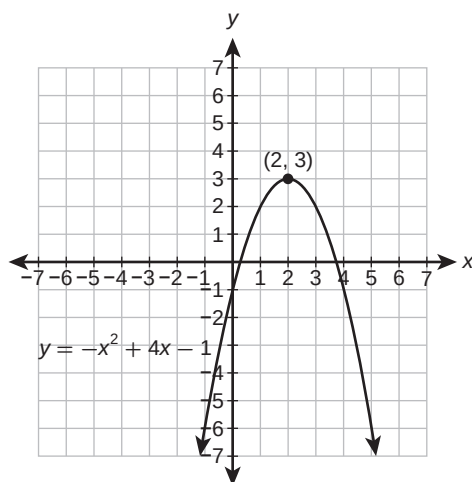
$a = -1$, $b = 4$, and $c = -1$. Since $a < 0$, the parabola opens down.

Use a and b to find the x -coordinate of the vertex, and then substitute to find the y -coordinate.

$$x = -\frac{b}{2a} = -\frac{4}{2 \cdot (-1)} = 2$$

$$y = -x^2 + 4x - 1 = -2^2 + 4 \cdot 2 - 1 = -4 + 8 - 1 = 3$$

The vertex is $(2, 3)$. Since the parabola opens down, the vertex is the highest point. The graph must cross the x -axis at two points. Therefore, the function has two real zeros. Sketch the graph to verify.



Graphing Quadratic Functions in Factored Form

Using the factored form of a quadratic function can help you graph the function.

Factored Form of a Quadratic Function

The factored form of a quadratic function is $f(x) = a(x - r_1)(x - r_2)$.

The graph of $f(x) = a(x - r_1)(x - r_2)$ is a parabola with these characteristics:

- It opens up if $a > 0$; it opens down if $a < 0$.
- It has x-intercepts r_1 and r_2 .
- It has an axis of symmetry with equation $x = \frac{r_1 + r_2}{2}$ (halfway between the x-intercepts).

► **Think About It** The x-intercepts of the graph of $f(x)$ are the roots of the equation $f(x) = 0$. By the zero product property, if $a(x - r_1)(x - r_2) = 0$, then $x - r_1 = 0$ or $x - r_2 = 0$. Solving these two equations for x gives the roots $x = r_1$ and $x = r_2$.

EXAMPLE 4

Use the factored form to graph the function.

A $y = x^2 - 2x - 3$

SOLUTION

Factor the trinomial.

$$y = x^2 - 2x - 3 = (x + 1)(x - 3)$$

Since $a = 1$, the parabola opens up. Since $r_1 = -1$ and $r_2 = 3$,

the x -intercepts are -1 and 3 . Since $\frac{r_1 + r_2}{2} = \frac{-1 + 3}{2} = \frac{2}{2} = 1$,

the equation of the axis of symmetry is $x = 1$. Since the axis of symmetry passes through the vertex, the x -coordinate of the vertex must be 1 .

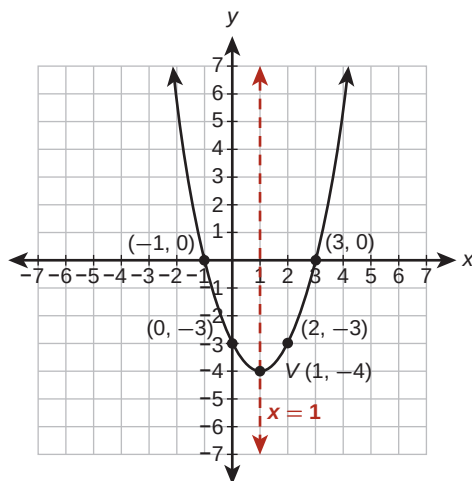
Substitute 1 for x in the function to find the y -coordinate of the vertex.

$$y = x^2 - 2x - 3 = 1^2 - 2 \cdot 1 - 3 = 1 - 2 - 3 = -4$$

The vertex, V , is $(1, -4)$.

Find two more points, such as $(0, -3)$ and its reflection, $(2, -3)$.

Then sketch the curve through all the points.



B $y = -0.4x^2 + 0.4x + 4.8$

SOLUTION

Factor out -0.4 and then factor the trinomial.

$$y = -0.4x^2 + 0.4x + 4.8 = -0.4(x^2 - x - 12) = -0.4(x - 4)(x + 3)$$

Since $a = -0.4$, the parabola opens down. Since $r_1 = 4$ and $r_2 = -3$,

the x -intercepts are 4 and -3 . Since $\frac{r_1 + r_2}{2} = \frac{4 + (-3)}{2} = \frac{1}{2} = 0.5$,

the equation of the axis of symmetry is $x = 0.5$. Since the axis of symmetry passes through the vertex, the x -coordinate of the vertex must be 0.5.

Substitute 0.5 for x in the function to find the y -coordinate of the vertex.

$$y = -0.4(x - 4)(x + 3) = -0.4(0.5 - 4)(0.5 + 3) = 4.9$$

The vertex, V , is $(0.5, 4.9)$.

Find two more points, such as $(2, 4)$ and its reflection, $(-1, 4)$. Then sketch the curve through all the points.

